

Python Development Internship Syllabus (Core Python + Web Development + APIs)

Duration: 14 Weeks Internship Program

Goal: Interns ko Python programming, automation, APIs aur backend development sikhana taki wo real-world applications aur backend systems develop kar saken.

Module 1: Introduction to Programming & Python

- Programming Basics
- Python Installation
- VS Code Setup
- Python Syntax
- Variables & Data Types

Assignment: Simple Python calculator

Module 2: Python Fundamentals

- Operators
- Conditions
- Loops
- Functions
- Input/Output
- Type Casting

Assignment: Student marks system

Module 3: Data Structures in Python

- Lists
- Tuples
- Sets
- Dictionaries
- Nested Data Structures

Assignment: Employee management system

Module 4: Advanced Python

- Modules & Packages
- File Handling
- Exception Handling
- Lambda Functions
- Map, Filter, Reduce

Assignment: File reader/writer system

Module 5: Object-Oriented Programming (OOP)

- Classes & Objects
- Constructors
- Inheritance
- Polymorphism
- Encapsulation

Assignment: Bank management system

Module 6: Python Libraries

- Datetime
- Random
- Math
- OS Module
- JSON Handling

Assignment: Utility tools project

Module 7: Database with Python

- SQLite Basics
- MySQL Connection
- CRUD Operations
- Database Queries

Assignment: Student database project

Module 8: APIs & Requests

- REST API Basics
- Requests Library
- GET & POST APIs
- JSON Response Handling

Assignment: Weather API project

Module 9: Web Development with Flask

- Flask Introduction
- Routes
- Templates
- Forms
- API Creation

Assignment: Flask CRUD app

Module 10: Django Basics

- Django Introduction
- Models
- Views
- Templates
- Admin Panel

Assignment: Django blog system

Module 11: Automation with Python

- Email Automation
- Excel Handling
- PDF Handling
- Web Scraping Basics

Assignment: Automation project

Module 12: Git & Deployment

- Git Basics
- GitHub
- Deployment Basics
- Python Hosting

Assignment: Deploy Python project

Module 13: Final Project

- Library Management System
- Inventory Management
- API-based Application
- Admin Dashboard

Final Submission with GitHub & Live Link

Daily Practice Structure

Time	Activity
1 Hour	Theory
3 Hours	Coding Practice
1 Hour	Assignments & Tasks
30 Minutes	Doubt Session

Tools & Technologies

- Visual Studio Code
- Python 3
- Flask
- Django
- MySQL
- SQLite
- Git & GitHub
- Postman

Internship Outcome

- Python programming efficiently kar payega
- APIs integrate aur create kar payega
- Flask/Django applications bana payega
- Database handle kar payega

- Automation scripts develop kar payega
- Live deployment kar payega
- Freelancing aur backend developer role ke liye ready hoga